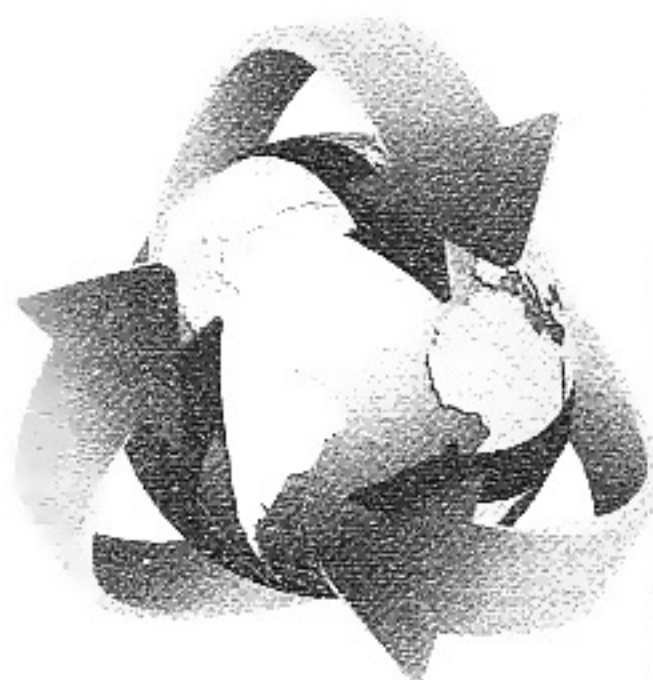


12 PEOPLE AND NATURE



REDUCE-REUSE-RECYCLE

Nature does nothing uselessly.

Aristotle

Sunshine is delicious, rain is refreshing, wind braces us up, snow is exhilarating; there is really no such thing as bad weather, only different kinds of good weather.

John Ruskin

□ VOCABULARY INPUT

Dealing with this topic you will come across the following vocabulary. Read it and look up the unknown words in your dictionary.

- **climate:** temperature, humidity, atmospheric pressure, wind, rainfall
- **weather:** bright and clear day, warm, hot (heat), sun shines, sunny intervals, sun rises/sets, rain (drizzle, light showers, it rains cats and dogs, it pours, light/heavy rain, it rains heavily, rainbow), thunderstorms, lightning, dew, breeze, cold wind blows, Indian summer, foggy, misty, muddy, damp, cold, cool, chilly, frosty (to freeze), icy, snowy, nasty, gloomy, unpleasant, piles of snow, thaw, slush, it sleets, it hails, temperature rises/drops
- **seasons:** *spring* (nature awakens, trees blossom, flowers bloom), *summer* (ripe fruit and vegetables are collected and bottled, harvest, crop) *autumn* (tree leaves turn yellow, trees shed leaves, flocks of birds move south) *winter* (lakes and rivers freeze, icicles hang from the roofs, snow melts and turns into slush, slippery roads)
- **environmental issues:**
 - pollution of water:* polluted water, factories, chemical waste, industrial and city waste
 - pollution of soil:* farmers, pesticides, fertilisers
 - pollution of air:* carbon dioxide, lead, radioactive particles, exhaust fumes, greenhouse effect
 - acid rain:* sulphuric acid in the clouds, infertile soil, destruction of tropical rain forests
 - destruction of the ozone layer:* aerosol sprays (CFCs chlorofluorocarbons)
 - global warming:* increased temperature, melting of mountain glaciers, the rising of the sea level, floods, precipitation, evaporation, intense rainstorms, dry soil
 - disappearance of animal and plant species:* The Red List of Threatened Species, extinction risk, in danger
 - natural disasters:* earthquakes, fire, droughts, volcanic eruptions, tornadoes, hurricanes, floods, tsunami waves, avalanches
 - devastation, damage, destroy, degradation, collapse, victims, survivors, rescuers, volunteers, humanitarian aid, pollution, starvation, famine, illiteracy, lack of food and energy sources
- **solutions:**
 - energy sources:* fossil fuels, nuclear energy, geothermal energy, solar energy, hydroelectric energy, tidal power and wind power
 - reduction waste model:* reduce – reuse – recycle, sewage treatment plants
 - education/organisations/government:* warnings, publicity, mass campaigns, clean air acts, water pollution acts and endangered species acts, Greenpeace, WWF (World Wildlife Fund), Friends of the Earth
 - forbidden/banned/not allowed to:* use DDT, phosphates in soaps, ozone-destroying chemicals in aerosol sprays, emissions of carbon dioxide and other greenhouse gases

□ DISCUSSION QUESTIONS

Before reading the text, discuss the following questions in class.

1. Why is the weather a good topic for conversation among people?
2. What kind of weather is ideal for summer holidays near the sea and for winter holidays in the mountains?
3. Which natural disasters have you recently heard about?
4. Which ecological issues concern our country?
5. What do you know about the activities of environmentalists?
6. What are the consequences of global warming?
7. What causes air, water and soil pollution?
8. What part can you play in "saving our planet"?
9. Which energy sources are more likely to be used in the future to a greater extent?
10. What does 'Reduce-Reuse-Recycle' mean to you? Explain.

□ TEXT

Read the texts below in silence. Check the meaning of the words in bold and answer the questions below.

CURRENT ECOLOGICAL ISSUES

The most serious ecological problems today are the **pollution** of air, water and **soil**, the destruction of ecosystems, the disappearance of many **species** of animals and plants, sudden changes in weather, **acid rain**, global warming, the destruction of tropical rain forests for firewood and building materials and the destruction of the **ozone layer**. Most of this **damage**, which is **accelerating** at a faster and faster **rate**, has not been caused by natural catastrophes but by man himself. Technical progress, which enables us to live a more comfortable life, leads on the other hand to the **degradation** of the quality of the environment in which we live. The growing number of cars **contributes** to air pollution. Factories produce large quantities of **carbon dioxide**, **lead** and **radioactive particles** like **exhaust fumes** which get into the air. Some of them in combination with water create **sulphuric acid** in the clouds. These dangerous clouds are then blown by the winds and later fall as acid rain which kills trees, then gets into the soil and makes it **infertile**. The polluted rain water is carried to the rivers and all life disappears from them. Factories directly **increase** water pollution because many of them are built near rivers into which they put their chemical **waste**. Farmers use large amounts of **pesticides** to kill certain **insects** that could be killed by other less dangerous **means**. **Fertilisers** also pollute the soil and may have negative effects on human health. Industrial and city waste, which are **dumped** near lakes and rivers, pollute the soil, water and air and **spoil** the natural scenery. If the destruction of the environment continues, then life on this planet will be **endangered**. Scientists warn that the average temperature of the atmosphere is increasing. There is a tendency towards global warming all over the world. It causes the **melting of mountain glaciers** and the **rising** of the sea level, which may cause the

flooding of coastal areas. According to scientists, the same trends will continue in the future.

1. Why is our planet in danger?
2. What are the most serious ecological problems mentioned in the text above?

NATURAL DISASTERS

Climate changes and their **consequences** directly and indirectly affect our lives. Many countries have recently been **hit** by different natural disasters: **earthquakes**, fire, **droughts**, volcanoes, tornadoes, hurricanes, **floods**, tsunami waves, etc. One of the deadliest disasters in modern history was the *Indian Ocean Earthquake*, known by the scientific community as the *Sumatra-Andaman Earthquake*. It **occurred** on December 26th in 2004. The earthquake and its tsunami killed more than 283,100 people and **devastated** coastal communities in Indonesia, Sri Lanka, India, Thailand and elsewhere. The Haiti earthquake on January 27th 2010 with a magnitude of 7 and at least 52 **after-shocks** caused major damage to the nearby capital and its surroundings. Approximately three million people were affected by the **quake**. The majority of **residences**, commercial buildings and notable **landmarks** either **collapsed** or were destroyed. Damage to communication systems, transport facilities and electrical networks caused initial relief problems for **rescuers**, doctors, engineers and other **humanitarian aid** supporters. Earthquake in Japan in 2011 was the most powerful earthquake in the world. It **triggered** enormously destructive tsunami waves of up to 40.5 metres which destructed a number of residences, infrastructure and caused the nuclear accidents.

3. Which natural disasters are mentioned in the text above? Which of them is the most dangerous and why?
4. What are the consequences of natural disasters such as tsunamis and earthquakes?

FOOD AND ENERGY SOURCES

In October 1999, the human population of the Earth **reached 6 billion**. We have to realise that we cannot expand forever because the Earth's **resources** are not **limitless**. In fact, we have passed our **sustainable** limits for both of our major food and energy sources and we are very quickly reaching our fresh water limits. Many organisations throughout the world warn of the dangers of **burning fossil fuels** and of nuclear waste **by-products**. They **promote** alternative energy sources that could be used without producing by-products and are **renewable**. After the Fukushima accident in 2011 in Japan, more and more people protest against nuclear power stations. Solar energy is used on a smaller **scale** for example to heat up swimming pools. It could be used on a larger scale to run cars or spaceships. Wind power, geothermal energy (**obtained** from the internal **heat** of the planet) or hydro-electricity will also be used more in the future. More than 1 billion people live in total **poverty** and many children die every year of **starvation**. There is a **remarkable** difference between certain parts of the world, such as Western Europe or the USA, with consumer societies and poor African states where people die of starvation.

5. Why do we need renewable sources of energy?
6. Which alternative sources of energy are discussed in the text above?

EXTINCTION OF SPECIES

Every organism needs certain living conditions, such as temperature, water, light and soil, to survive. Nowadays, the existence of some plants and animals is in danger. *The Red List of Threatened Species*, created in 1963, gives **precise** criteria to **evaluate** the extinction risk and **provides** a list of species that are in danger. For example elephants, tigers, pandas, whales, corals, great apes, **marine turtles**, polar bears, rhinos, snow leopards and dolphins are at the top. Worldwide organisations, such as Greenpeace, WWF (World Wildlife Fund), Friends of Earth and others, are trying to protect endangered plant and animal species. The most endangered living animal species in Slovakia are **chamois**, **marmot** and **lynx** and from plant species the most endangered is **edelweiss**.

7. Which plant and animal species are endangered? Name some of them.

ECOLOGY AWARENESS IN SLOVAKIA

Many of these problems also concern our country. Sudden weather changes often lead to high water levels in many areas of Slovakia. The most devastating was the flood in spring 2010, which caused enormous damage. Our country has to follow certain EU **measures** concerning the environment. The *Kyoto Protocol* is an international **agreement** linked to the *United Nations Framework* called *Convention on Climate Change*. The countries which have **ratified** it, **commit** themselves to reducing their emissions of carbon dioxide and five other **greenhouse gases**. Another major environmental issue are the **nuclear power plants** in Jaslovské Bohunice and Mochovce. As the older reactors in Bohunice did not meet the European standards, they were shut down in 2006 and 2008. However, the two new reactors in Mochovce are **due to** be completed in 2013 and 2014.

8. Which ecological issues in Slovakia are dealt with in the passage above?

SOLUTIONS AND SUGGESTIONS

Man certainly has the intelligence and the means to solve ecological problems and to save this planet for future generations. Environmental organisations, such as *Greenpeace*, **point to** the most serious ecological problems and try to give their **warnings** the greatest publicity possible through the mass media and public campaigns. Many of their activities have been successful and the growing **public attention** has made governments **pass** various **laws aimed at** improving the environment. For example, the use of DDT was **banned by law** all over the world, **catalytic converters** and **unleaded petrol** in cars are compulsory in many countries, the use of **phosphates** in soaps is forbidden, and most industrial countries have agreed to stop using ozone-destroying chemicals in

aerosol sprays. **Concern** for the environment should be a part of people's lives. Therefore, there is a great need for environmental education. Children of all ages should be educated to have a much better understanding of the fact that this planet, weather, climate, water, air, soil, people, animals, birds and plants create one ecosystem and cannot exist separately.

9. Which laws were passed in order to improve the environment?

10. Suggest possible solutions for solving some ecological problems using the ideas from the text.

□ VOCABULARY LIST

to accelerate – urýchľovať
acid rain – kyslý dážď
aftershock – následný otras
agreement – zmluva, dohoda
billion – miliarda
by-product – vedľajší produkt
carbon dioxide [ka:ɪbən daɪ'ɒksaɪd] – oxid uhličitý
catalytic convertor – katalyzátor
chamoise [ʃæmwa:] – kamzík
coastal – pobrežný
to collapse – zrútiť sa
to commit – zaviazat' sa
concern – záujem
to contribute – prispievať
consequence [kɒnsɪkwəns] – následok
damage – škoda, ujma
degradation – zhoršenie
to devastate – spustožiť
drought [draʊt] – obdobie sucha
to dump – vysypať, skladať odpad
earthquake – zemetrasenie
edelweiss – plesnivec
to endanger – ohroziť
to evaluate – ohodnotiť
exhaust fumes [ɪg'zɔ:st fju:mz] – výfukové plyny
fertiliser (AmE: fertilizer) [fɜ:təlaɪzə(r)] – hnojivo
flood – povodeň
flooding – zatopenie
framework – rámec
greenhouse gas – skleníkový plyn
heat – horúčava
to hit – udrieť, zasiahnuť
humanitarian aid – humanitárna pomoc

to increase – zvýšiť
infertile [ɪn'fɜ:taɪl] – neúrodná
insect – hmyz
landmark – pamiatka, pamätihodnosť
lead [led] – olovo
limitless – neobmedzený
lynx [lɪŋks] – rys
marine turtle [məri:n] – morská korytnačka
marmot [ma:mət] – svišť
means – prostriedok
measure – opatrenie
melting of mountain glaciers [glæsiə(r)s] – topenie horských ľadovcov
nuclear power plant – jadrová elektráreň
to obtain – získať
to occur [ə'kɜ:(r)] – vyskytovať sa
ozone layer [leɪə(r)] – ozónová vrstva
pesticide [pestɪsaɪd] – pesticíd (na ničenie hmyzu)
phosphate [fɒsfeɪt] – fosfát
to point to – poukazovať na
pollution [pə'lu:ʃn] – znečistenie
poverty – chudoba
precise [pri'saɪz] – presný
to promote – propagovať
to provide – poskytnúť
public attention – pozornosť verejnosti
(earth)quake – zemetrasenie
radioactive particle – rádioaktívna častica
rate – miera
to ratify [ræɪfaɪ] – ratifikovať
to reach – dosiahnuť
remarkable – pozoruhodný
renewable – obnoviteľný
rescuer – záchranár
residence – obydlie

resource – zdroj
rising – stúpajúci
scale – rozsah
soil – pôda
species [spi:ʃi:z] – druhy
to spoil – pokaziť, zničiť
starvation – hladovanie
sulphuric acid [sʌlf rɪk æsɪd] – kyselina
 sírová
sustainable [sə'steɪnəbl] – udržateľný

threatened [θretənd] – ohrozený
to trigger – spustiť
unleaded petrol – bezolovnatý benzín
warning – varovanie
waste – odpad

to be banned by law – zákonom zakázaný
to burn fossil fuels – spaľovať pevné palivá
to pass laws aimed at – schváliť zákony
 zamerané na

□ VOCABULARY PRACTICE

Task 1 Fill in the following words in their correct form in the following passages about seasons.



BLOOM/BLOSSOM/MELT/SET/PLANT

Spring begins on March 21st. Nature awakens, snow ¹_____ and the temperature rises. Trees ²_____ and flowers, such as snowdrops, daffodils and tulips, ³_____. People enjoy working in their gardens: they dig the soil and ⁴_____ seeds. Light showers and sunny intervals are typical for this season. Days become longer because the sun ⁵_____ later.

FLOOD/DRIZZLE/DEW/THUNDERSTORM/HARVESTING

Summer begins on June 21st. The weather is great for nature lovers, hikers, campers and beach lovers. Ripe fruit are collected and bottled. This is also the season of collecting mushrooms and ⁶_____. Summer heat alternates with ⁷_____, showers, downpours, rain, hail and ⁸_____ with lightening. When the sky clears up and it is a bright and clear day again, a beautiful rainbow can be seen in the sky. In the morning, there is often ⁹_____ on the grass. Weather extremes, such as droughts and ¹⁰_____, can often spoil all the year's crops.

RISE/FLY/SLEET/SHED/TURN/PICK

Autumn starts on September 23rd. Days get shorter and the sun ¹¹_____ sooner and sets earlier. Indian summer brings the last warm days of autumn. Children ¹²_____ their kites, while their parents clean their gardens and ¹³_____ apples, pears, cabbages, cauliflowers and potatoes. Picking mushrooms, collecting walnuts, rosehips and

chestnuts are popular family activities. This is also the season of hunting. The weather is unpredictable and nature changes a lot. It is foggy, misty, muddy and damp. It rains cats and dogs, it ¹⁴ _____ and hails. People wear raincoats and umbrellas in order not to get soaked. Children wear Wellington boots and jump into puddles on purpose. Tree leaves ¹⁵ _____ yellow, red and brown and then the trees ¹⁶ _____ them. Flocks of birds move to southern countries.

FROSTY/ICICLES/SLUSH/ICY/NASTY

Winter begins on December 21st. The temperature drops rapidly and the lowest temperature can go as low as twenty degrees below zero. It is cold, snowy, ¹⁷ _____ and icy. Children build snowmen and throw snowballs. They enjoy skiing, sledging and skating on ice rinks or frozen ponds. ¹⁸ _____ hang down from the roofs. Drifts of snow and ¹⁹ _____ roads often cause traffic problems. It is ²⁰ _____ when the northern wind blows and the thaw begins. The ice and snow melts and turns into ²¹ _____.

(Adapted from <http://en.wikipedia.org>)

Task 2 Add the sentences in column A to the most likely responses in column B.



A

- Example:* It's a lovely day, isn't it? a
1. It's getting warm, isn't it? _____
 2. Nasty day, isn't it? _____
 3. Do you think that this sunny weather will last? _____
 4. It's going to freeze at night. _____
 5. It's a gloomy day. _____
 6. What's the weather going to be tomorrow? _____
 7. What's the weather report for today? _____
 8. We're having a hot spring this year, aren't we? _____
 9. The heat's unpleasant, don't you think? _____
 10. It looks like rain, doesn't it? _____

B

- a) Yes, it is.
- b) Yes, but it seems to be clearing up.
- c) Oh, yes. I hate it when it's pouring with rain.
- d) I hope it'll keep fine.
- e) It'll be cold and windy.
- f) I'm afraid it does.
- g) I hope not.
- h) They said light showers in the west with mild wind.
- i) Oh, yes, very unpleasant.
- j) Oh, yes. But it's going to get cooler again.
- k) Yes, this weather is quite unusual for this time of year, don't you think?

Task 10 Match the words in column A to the words in column B to make pairs.

8

A

Example: endangered

1. acid _____
2. ozone _____
3. oil _____
4. aerosol _____
5. greenhouse _____
6. solar _____
7. global _____
8. fossil _____
9. catalytic _____
10. water _____

B

- a) energy
- b) pollution
- c) spray
- d) fuels
- e) layer
- f) converters
- g) species
- h) warming
- i) rig
- j) effect
- k) rain

Task 11 What should be done about the following things to save our planet?**Choose one of the words to complete the following sentences.**

8

- The use of DDT was _____ by law all over the world.
A) *banned* B) *allowed* C) *reduced*
- Catalytic converters and unleaded petrol in cars are _____ in many countries.
A) *important* B) *compulsory* C) *polluted*
- The use of phosphates in soaps is _____ .
A) *forgotten* B) *forgiven* C) *forbidden*
- Most industrial countries have agreed to _____ using ozone-destroying chemicals in aerosol sprays.
A) *cancel* B) *stop* C) *abolish*
- Fossil or traditional fuels should be _____ by alternative forms of energy, such as solar energy, wind power, geothermal energy or hydroelectricity.
A) *replaced* B) *reduced* C) *recovered*
- Factories should be _____ for putting their chemical waste into rivers.
A) *concerned* B) *fined* C) *warned*